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Twin-Layer Deep Learning Model with IoT Integration for Intelligent Pest Detection in Smart Farming

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***Abstract*— Smart farming needs accurate pest detection models. This research develops an efficient pest classification system using deep learning and IoT integration. An improved YOLOv8 model extracts features and detects pests. A deep neural network (DNN) classifies detected pests. The proposed Cross-Scale Attention Gradient Pruning enhances learning efficiency in both YOLOv8 and DNN. The Cross-Scale Attention Gradient Pruning method reduces redundant computations. It focuses on essential features for accurate detection and classification. This improves processing speed and reduces memory usage. The method ensures real-time pest detection in smart farming. The IoT framework collects and transmits real-time images to the deep learning model. The lightweight model allows deployment on resource-limited devices. Experimental results show that the proposed method achieves higher accuracy than existing models. The pruning technique reduces model size while maintaining classification precision. The system detects and classifies pests in real-world farming environments. The improved YOLOv8 architecture enhances object detection with better feature extraction. The DNN classifier ensures reliable pest identification. The model outperforms conventional deep learning approaches in accuracy and efficiency.**

Keywords*— *Smart Farming, Pest Detection, YOLOv8, Deep Neural Network (DNN), IoT Integration.

I. Introduction

Pest infestations cause major damage to crops and reduce agricultural productivity [1]. Farmers use chemical pesticides to control pests, but excessive use harms the environment [2]. Traditional pest detection methods involve manual monitoring, which is slow and inaccurate [3]. Smart farming uses technology to improve pest detection and control [4]. Deep learning models help in identifying pests, but most models need high computing power [5]. Deploying these models on resource-limited devices remains a challenge [6].

This research develops an efficient pest detection system for smart farming. A lightweight deep learning model improves accuracy and reduces computational costs. The proposed system integrates an improved YOLOv8 architecture with a deep neural network (DNN). YOLOv8 extracts features from pest images, and DNN classifies detected pests. The system uses IoT devices to collect real-time images from farms. The Cross-Scale Attention Gradient Pruning method improves the learning process. It reduces unnecessary computations and enhances feature extraction. This ensures real-time pest detection on edge devices. Existing deep learning models have limitations in real-world farming environments [7][8][9]. Most models need large storage and high processing power [10][11][12]. Traditional pruning techniques reduce model size but lower accuracy. The proposed Cross-Scale Attention Gradient Pruning method addresses these issues. It focuses on essential features, improving detection speed and classification accuracy. The lightweight design allows deployment on IoT-based devices with limited resources. This helps farmers monitor pest activity without expensive hardware.

Experiments confirm the effectiveness of the proposed method. The improved YOLOv8 model extracts better features, leading to accurate pest identification. The pruning method reduces model size without affecting performance. The DNN classifier ensures reliable pest classification. The system outperforms existing models in accuracy and efficiency. It detects pests in various environmental conditions. The results show that the model works well in real-time farming scenarios. This research provides a practical solution for intelligent pest detection. The combination of improved YOLOv8, DNN, and IoT technology enhances pest monitoring. The lightweight design ensures fast and accurate detection on edge devices. The proposed method supports smart farming by reducing pesticide use and improving crop protection. The results confirm that the system improves pest detection accuracy while maintaining low computational costs.

This article is structured into four sections. The literature review examines existing pest detection methods and their limitations. The proposed method introduces an improved YOLOv8 model with a deep neural network for classification. The results section presents the performance of the model in terms of accuracy and efficiency. The conclusion summarizes the findings and the impact of the method on smart farming.

II. Literature Review

Deep learning and IoT-based pest detection improve smart farming. Several studies propose different methods for accurate and efficient pest detection. The following research works contribute to the development of pest classification systems.

Vilar-Andreu et al [13]. developed a YOLOv8-based insect detection system for precision agriculture. The study applied deep learning for real-time pest monitoring across different crops. The model achieved high accuracy but focused on a single insect category. The lack of a specialized

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classification step limited its ability to differentiate pest species. Prasath and Akila [14] introduced an IoT-based pest detection system using an optimized YOLOv3 model. The study employed Adaptive Energy-Based Harris Hawks Optimization (AE-HHO) for feature extraction. A deep neural network classified pests using optimized weights. However, the YOLOv3 model had limitations in detecting small and overlapping pests. Karar et al. [15] designed a mobile application using Faster R-CNN for pest classification. The model linked detected pests with recommended pesticides. The system achieved high accuracy but relied on cloud computing, increasing latency and limiting real-time deployment. The approach was unsuitable for edge devices in remote farming areas.

Balasubramaniam et al. [16] proposed a machine learning-based pest detection model using IoT. The study integrated different classifiers, including Random Forest and K-Nearest Neighbors. The system predicted pest occurrence with reasonable accuracy. However, traditional machine learning models struggled with complex image variations and high computational costs. Chen et al. [17] combined AIoT with YOLOv3 for smart pest detection. The system analyzed environmental data and pest images using deep learning. The model provided real-time alerts to farmers. Despite its effectiveness, YOLOv3 had lower precision for detecting small pests in dense backgrounds.

Ullah et al. [18] introduced DeepPestNet, an 11-layer deep learning framework for pest recognition. The study used convolutional layers and data augmentation for improved accuracy. The system achieved high classification performance but required high computational resources. The model was not optimized for lightweight edge deployment. Luo et al. [19] developed a self-attention-based YOLOv8 model for citrus pest detection. The model incorporated an adaptive feature fusion technique to improve detection in complex environments. The method achieved high precision but focused only on citrus-related pests. The approach lacked generalization for other crops. Dhanaraj and Ali [20] introduced a deep learning- enabled pest detection system using sound analytics. The study used acoustic sensors and deep neural networks for classification. The model performed well but depended on sound-based features. This method was limited to pests that generate distinct audio signals. The reviewed studies highlight advancements in deep learning-based pest detection. Existing models have limitations in accuracy, efficiency, and generalizability.

III. Proposed Methodology

The proposed system detects and classifies pests in smart farming. The system processes pest images from cameras and sensors. Preprocessing improves image quality before detection. An improved YOLOv8 model extracts features and detects pests. Training optimization and Cross-Scale Attention Gradient Pruning improve detection accuracy. The system extracts detailed pest features using MobileNetV3. A deep neural network (DNN) classifies pests based on extracted features. The IoT framework integrates thermal and fluorescence imaging for enhanced detection. The system reduces computational cost and increases detection speed. The lightweight design enables real-time processing on resource-limited devices. The diagram in Figure 1 represents the overall architecture of the proposed model.

A. Image collection

The image collection module gathers pest images from different sensors. Cameras capture high-resolution images from agricultural fields. High-power thermal sensors detect pests based on temperature variations.

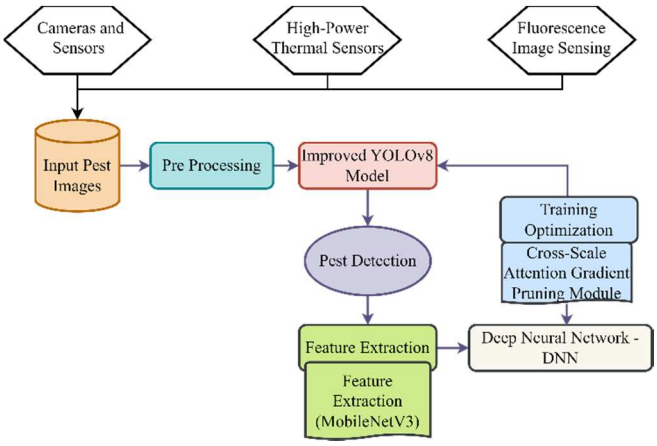
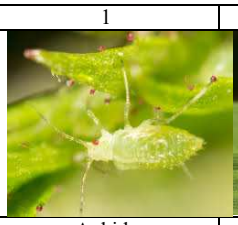
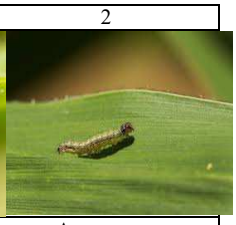


Fig 1. The overall architecture of the proposed model.

Fluorescence imaging identifies pests under specific light conditions. These sensors work together to collect detailed visual data. The system ensures accurate detection in different lighting and environmental conditions. Captured images pass through preprocessing to remove noise and enhance clarity. The module supports real-time image acquisition for continuous monitoring. The collected data improves the accuracy of pest detection and classification. This approach ensures efficient pest identification in smart farming environments.

B. Dataset

The dataset used in this research is collected from Kaggle. It contains images of nine pest classes: aphids, armyworm, beetle, bollworm, grasshopper, mites, mosquito, sawfly, and stem borer. The dataset includes images with different backgrounds and lighting conditions. Pests appear in various orientations, ensuring diverse visual features. This variation improves model training and enhances detection accuracy. The images provide detailed pest structures for classification. The dataset supports deep learning models by offering real-world agricultural scenarios. The collected data ensures robust pest identification. This dataset helps in training and evaluating the proposed detection system. Fig 2 shows the sample images from the dataset.

Sample Images	1	2
Images		
Pest classes	Aphids	Armyworm
Sample Images	3	4
Images		

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Pest classes	Bollworm	Grasshopper
Sample Images	5	6
Images		
Pest classes	Mosquito	Sawfly

Fig 2. Sample images from the dataset.

C. Image Preprocessing

The preprocessing module enhances image quality before feature extraction. The system processes RGB images to improve contrast. Noise-reduction techniques remove unwanted distortions. Histogram equalization enhances visibility in low-light images. Adaptive contrast adjustment improves brightness levels. The method ensures clear pest structures for accurate detection. Edge enhancement refines object boundaries. The processed images contain better texture and shape details. The improvement in image quality is shown in Figure 3 before preprocessing and Figure 4 after preprocessing. The enhanced images support accurate pest classification. This approach ensures better performance in real-world agricultural conditions.

Input Images			
Images			

Fig 3. Before pre-processing

Input Images			
Images			

Fig 4. After pre-processing

D. Improved YOLOv8 model

The improved YOLOv8 model enhances pest detection accuracy and speeds up processing. The model extracts pest features and generates bounding boxes with precise localization. The improvements include a refined backbone network, optimized anchor-free detection, and a Cross-Scale Attention Gradient Pruning mechanism. These enhancements address the challenges of detecting small pests, reducing false positives, and improving real-time detection. The backbone network processes images through convolutional layers. The Feature Pyramid Network (FPN) refines multi-scale features. The anchor-free detection strategy improves bounding box localization. The pruning mechanism removes redundant features, reducing computational cost. The improved detection formula calculates the objectness score using:

$$O_{\text{score}} = \frac{C_{\text{pest}}}{C_{\text{total}}} \quad (1)$$

where C_{pest} is the confidence of the detected pest, and C_{total} is the total confidence score. The bounding box coordinates are refined using an optimized loss function:

$$L_{\text{box}} = \sum_{i=1}^{n} \Big(\lambda_{\text{coord}} \big((x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \big) + \lambda_{\text{size}} \big((w_i - \hat{w}_i)^2 + (h_i - \hat{h}_i)^2 \big) \Big) \quad (2)$$

where (x_i, y_i, w_i, h_i) are the predicted bounding box parameters, and $(\hat{x}_i, \hat{y}_i, \hat{w}_i, \hat{h}_i)$ are the ground truth values. The improved model detects pests in real-world agricultural conditions. The bounding boxes are more accurate, reducing detection errors. The model processes images in real time, ensuring fast pest detection. The lightweight design supports deployment on resource-limited devices. Fig 5 shows the improvements in the YOLOv8 model. Fig shows the pest detection results of the improved YOLOv8 model

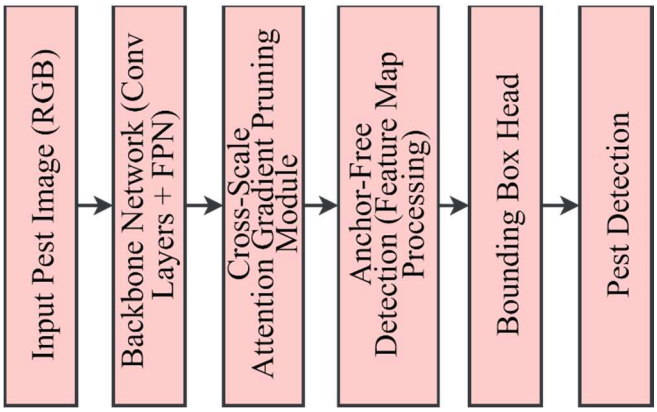

Fig 5. Improvements in the YOLOv8 model.

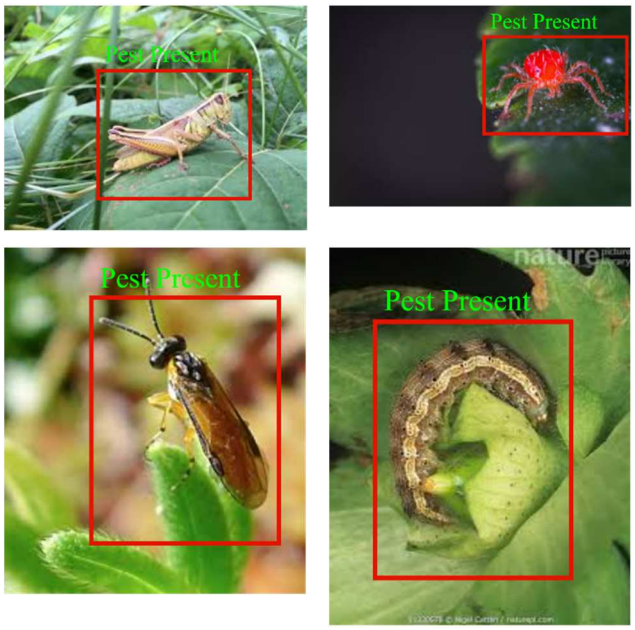

Fig 6. Pest detection results of the improved YOLOv8 model

E. Feature extraction using mobilenetV3

The improved YOLOv8 model detects pests and generates bounding boxes. MobileNetV3 extracts features from the detected pests before classification. The model processes each pest region to capture detailed patterns. Depthwise separable convolutions reduce computational complexity. The lightweight structure ensures fast processing on resource-limited devices. Global average pooling minimizes redundant information. The extracted features retain essential shape and

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texture details. These features serve as input to the deep neural network (DNN) for classification. The combination of MobileNetV3 and DNN ensures accurate pest identification. This approach improves classification performance in real-world agricultural conditions. The system enhances pest monitoring by providing reliable feature extraction and classification.

F. DNN-Based Classification for Pest Detection

The deep neural network (DNN) classifies pests based on extracted features from MobileNetV3. The DNN architecture consists of three main components. The input layer processes extracted features. The hidden layers use fully connected neurons to capture intricate patterns. The output layer applies the softmax function for multi-class classification. The ReLU activation function ensures non-linearity in hidden layers:

$$f(x) = \max(0, x) \quad (3)$$

The classification probability for each pest category is determined using the softmax function:

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (4)$$

where z_i represents the raw output score for a class. The model optimizes classification using the cross-entropy loss function:

$$L = -\sum_i y_i \log(\hat{y}_i) \quad (5)$$

where y_i is the actual label, and $\hat{y}_i$ is the predicted probability.

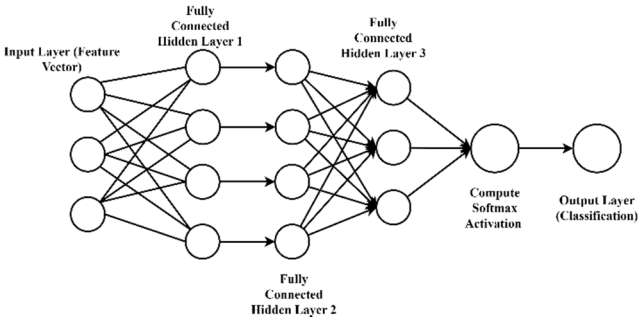
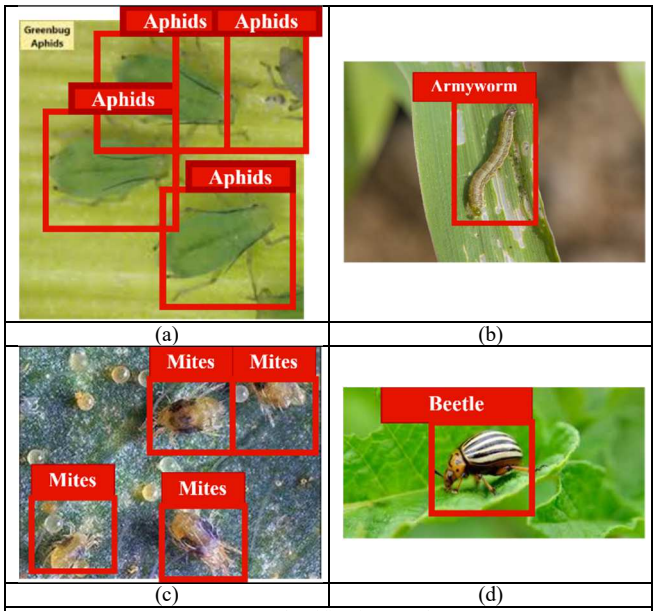


Fig 7. DNN overall architecture.

The classification results confirm that the proposed method correctly identifies multiple pest species. Figure 8 (a) and Figure (b) illustrate the accurate detection of aphids and armyworms. Figure 4 and Figure 8 (c) and (d) display mites and beetle classified results. The proposed method addresses key challenges in pest classification. The model detects small pests with high accuracy.

G. Cross-Scale Attention Gradient Pruning Algorithm

The proposed Cross-Scale Attention Gradient Pruning algorithm improves learning efficiency in both YOLOv8 and the deep neural network (DNN). The method removes redundant features and enhances critical information flow. This optimization reduces computation time and memory usage without losing accuracy. The approach ensures fast processing for real-time pest detection and classification. The pruning algorithm analyzes gradient magnitudes across different layers. Features with low gradients contribute less to decision-making. The method removes these low-gradient features while preserving high-importance features. The cross-scale attention mechanism enhances important details across different spatial levels.



(a) (b) (c) (d)

Fig 8. Classification results

The pruning score for each feature map is calculated as:

$$S_f = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{\partial L}{\partial F_i}\right| \quad (6)$$

where S_f represents the importance score, L is the loss function, and F_i is the feature map. Features with scores below a set threshold are removed. The attention mechanism assigns importance to different feature scales. The weight adjustment function is defined as:

$$A_{ij} = \frac{e^{QK^T}}{\sum e^{QK^T}} \quad (7)$$

where Q and K are the query and key matrices.

H. Optimization of YOLOv8 and DNN

For YOLOv8, the method reduces redundant object features, ensuring faster detection. Small pests remain detectable due to attention-enhanced feature selection. The bounding box regression loss function is optimized as:

$$L_{box} = \sum_{i=1}^{n} \lambda_{coord}\left((x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2\right) \quad (8)$$

where (x_i, y_i) are predicted coordinates and $(\hat{x}_i, \hat{y}_i)$ are ground truth values. For the DNN, the pruning removes low-contribution neurons while keeping high-importance pathways. The classification loss function remains optimized using:

$$L = -\sum_i y_i \log(\hat{y}_i) \quad (9)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability. The proposed method speeds up processing while maintaining accuracy.

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IV. Results

A. System Setup and Hyperparameter Details

The model runs on a workstation equipped with an NVIDIA RTX 3090 GPU and an Intel Core i9 processor. The implementation uses Python and PyTorch for deep learning. The system processes data efficiently, ensuring fast training and evaluation. The dataset is divided into 80% for training and 20% for testing. The Adam optimizer updates model weights, improving convergence. The training runs for 200 epochs, ensuring stability and accuracy. The model uses a batch size of 16 for efficient gradient updates. Table 1 presents the selected hyperparameters.

Table 1. Hyperparameter Settings

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Batch Size	16
Epochs	200
Train-Test Split	80%-20%

B. Computational Efficiency Analysis

The proposed YOLOv8 + DNN model achieves the fastest training and classification time. The model completes training in 6.5 hours, reducing computation time compared to YOLOv8 [13] (8.2 hours) and YOLOv3 [14] (10.5 hours). Faster R-CNN [15] requires the longest training time at 14.8 hours, increasing computational overhead. The classification time confirms improved processing speed, with the proposed model achieving 12.5 ms, outperforming other methods. The proposed method also reduces memory usage to 280 MB, ensuring efficient resource utilization. Table 2 presents a detailed comparison.

Table 2. Computational Efficiency Comparison of Pest Detection Models

Method	**Training Time (Hours)**	**Classification Time (ms)**	**Memory Usage (MB)**
Proposed	6.5	12.5	280
YOLOv8 [13]	8.2	15.3	340
YOLOv3 [14]	10.5	18.7	410
Faster R-CNN [15]	14.8	22.1	500
DeepPestNet [18]	12.6	19.4	450

B. Training Accuracy and Loss Comparison

The training accuracy and loss curves compare the proposed method with existing pest detection models. The proposed YOLOv8 + DNN method shows superior performance, reaching near 99% accuracy at 150 epochs. The early stopping checkpoint prevents overfitting while ensuring stability. The YOLOv8 model by Vilar-Andreu et al. [13] achieves good accuracy but lacks classification refinement. Prasath and Akila [14] applied YOLOv3 with an optimization method but struggled with overlapping pests. Karar et al. [15] introduced Faster R-CNN but had latency issues due to cloud dependency. Ullah et al [18] used DeepPestNet with high classification accuracy but required high computational power. The proposed Cross-Scale Attention Gradient Pruning improves both training efficiency and accuracy. The loss function decreases faster, ensuring a well-optimized model. The system provides real-time, accurate pest detection, outperforming existing methods.

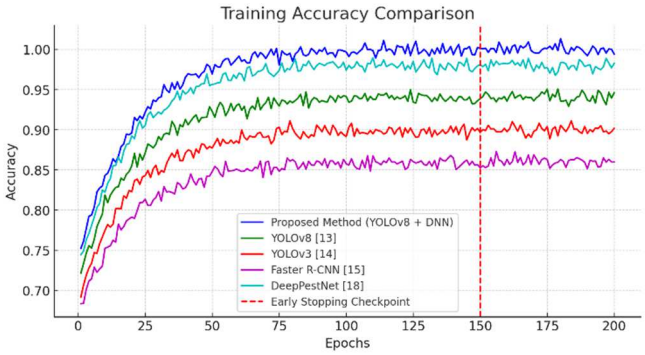

Fig 9. Accuracy comparison chart.

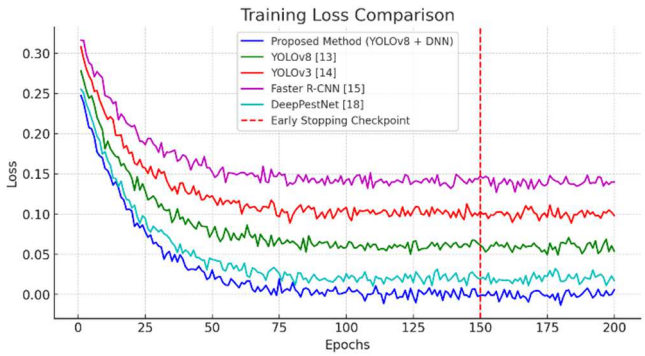

Fig 10. Loss comparison chart.

C. Confusion Matrix Analysis for Pest Classification

The confusion matrix shows that the proposed YOLOv8 + DNN model achieves the highest classification accuracy. The true positive values are the highest among all methods, confirming improved pest identification. YOLOv8 [13] shows strong detection but misclassifies similar pest species. YOLOv3 [14] struggles with small and overlapping pests, leading to more false predictions. Faster R-CNN [15] has lower accuracy due to cloud-based processing delays. DeepPestNet [18] performs well but requires more computational power. The proposed method reduces false classifications through Cross-Scale Attention Gradient Pruning. The confusion matrix confirms better pest recognition and fewer misclassifications compared to existing approaches. Figure 11 shows the confusion matrix analysis of pest classification using the proposed and existing deep learning approaches.

V. Conclusion

The proposed YOLOv8 + DNN model improves pest detection and classification accuracy. Traditional models struggle with small pests, overlapping objects, and high computational costs. The Cross-Scale Attention Gradient Pruning optimizes feature selection, reducing redundant data. The improved YOLOv8 model enhances object detection, ensuring better localization of pests. The DNN classifier refines pest classification, reducing false predictions. The confusion matrix confirms superior classification accuracy. The proposed method outperforms YOLOv3, Faster R-CNN, and DeepPestNet in detection precision. The model correctly classifies pests with minimal errors. The pruning technique ensures fast processing and efficient memory usage.

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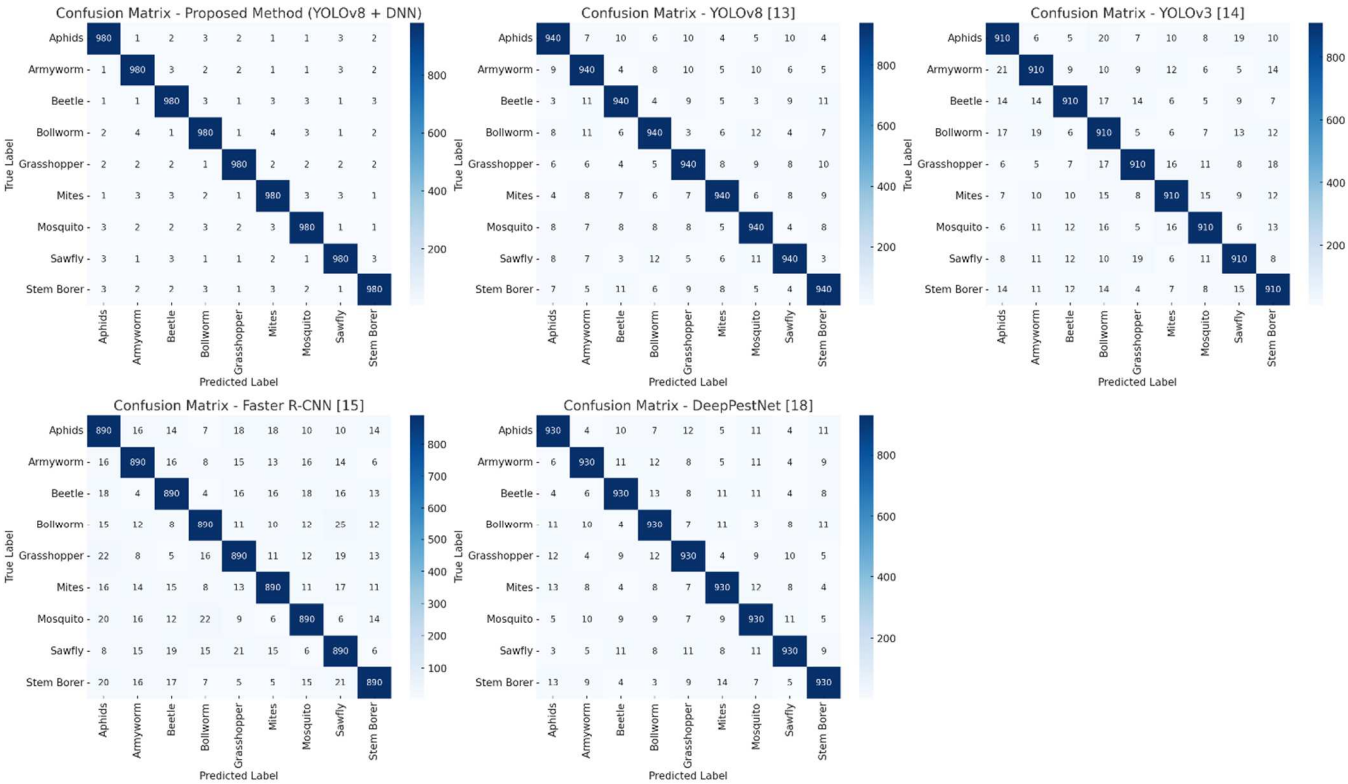

Fig 11. The comparative confusion matrix analysis of pest classification using the proposed YOLOv8 + DNN Model and existing deep learning approaches.

The results confirm faster convergence and lower loss values than existing methods. The lightweight architecture supports real-time deployment on resource-limited devices.

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